



AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER



Test Report No. 2023 – 0065



Agricultural Machinery – Biomass Shredder RET MPS 7.5HP

Test Report Number	2023 – 0065
Date of Issuance	March 7, 2023
Valid Until	March 7, 2028
Agricultural Machinery	Biomass Shredder
Type	Horizontal shaft, Rotating blade, Open cylinder, Shredding and chipping on the same chamber, Engine-driven, Mobile (Cart type)
Brand	RET
Model	MPS 7.5HP
Test Requested by	RETJR ENTERPRISE San Jose, Batangas

SUMMARY

Performance Criteria	Requirement as per PNS/PAES 244:2010	AMTEC Test
Input Capacity, kg/h	1000	1115.59
Shredding Efficiency, %, minimum	90	94.35
Noise Level, dB(A), maximum	96	94.1

^a As per manufacturer's specification/Not specified by the requesting party

Note: ND – No Data

OBSERVATIONS

1. Operator's manual (Based on PAES102:2000)	
a. Language	English
b. Safety and warnings	Provided
c. Operating information	Provided
d. Accessories and attachments	Not provided
e. Maintenance instruction	Provided
f. Storage	Provided
g. Handling, reception, transportation, assembly, and installation	Not provided
h. Specifications	Not provided
i. Dismantling and disposal	Not provided
j. Warranty	Not provided
k. Parts list	Not provided
2. Previous test reports of similar brand and model	None
3. Manufacturer	RETJR ENTERPRISE
4. Prime mover used	5.59 kW KRESS KE207 Gasoline Engine
5. Number of operators	Three (One feeding operator and two assistants for unclogging the output chute)
6. Loading of input	The loading hopper is accessible to the operator. Input materials were loaded individually into the hopper.
7. Adjusting and repairing of parts	Clearance of blades are not adjustable. Parts are accessible for repair and locally manufactured parts can be replaced.
8. Collecting of output	Output fell on the ground and were moved aside using rakes to avoid clogging of the output chute area. Shredded output was manually collected after the test operation.
9. Cleaning of parts	All parts are accessible for cleaning.
10. Transporting the machine	The machine is provided with cartwheels and towbar for mobility.
11. Safety features	The transmission assembly is provided with safety guard. Rubber guards were provided at the input and output chutes.
12. Failure or abnormalities that may be observed on the machine or its component parts during and after the operation	None
13. Steel bars, metal sheet or plate and heavy-duty mild steel shall be generally used for the different components of the machine	Steel plate and tool steel were used in the different components of the machine.

OBSERVATIONS (Cont'n)

14. Provided with output chute	Provided
15. Knives or cutting mechanisms shall be made of high carbon steel and/or tool steel	The blades are made up of carbon steel (see XRF Analysis at Table 1).
16. For open cylinder type of shredder, the angle of inclination with respect to horizontal axis for blades or cutting mechanism attached to the cylinder should be between 15° to 20° to facilitate conveyance and to avoid clogging of shredded materials inside the shredding chamber.	Not measured
17. Clearance between rotating blades and counter blades of the shredding chamber must be at least 6 mm.	The measured clearance between rotating blades and shredding chamber is 13.12 mm. There were no counter blades provided.
18. Cover for shredding blades must be at least 2.5 mm M.S. plate and/or B.I. plate or its equivalent material	The shredding chamber is made of 2.75 mm thick AISI 4140 (see XRF Analysis at Table 2).
19. Bolts and screws shall conform to the requirements of PAES 311 and 313	Not verified
20. Sizes of the parts of the machine shall be based on the specifications of the manufacturer	See Specifications Table in page 7 and 8.
21. The base shall be rigid and its rotating components shall be statically and dynamically balanced for stable operation	The base was rigid. All rotating parts were observed to be stable and balanced.
22. Machine shall be free from manufacturing defects	Based on ocular inspections, no manufacturing defects were observed during the test operation.
23. Cutting mechanism shall be adjustable, replaceable and can be sharpened	The blades can be adjusted, replaced, and sharpened.
24. All surfaces shall be free from rust and shall be coated with a suitable paint material	All non-rotating parts were painted and free of rust.
25. Belt cover or guard and provisions for belt tightening and adjustments shall be provided	A belt guard for the belt and pulley transmission system was provided. The machine was provided with belt tensioner.
26. Mechanism for immediate load disengagement between prime mover and biomass shredder shall be provided	Belt tensioner was provided to disengage the engine from the machine.
27. All welded parts shall be water-tight and smoothly polished and it shall pass visual inspection criteria for discontinuity of materials	Not verified

OBSERVATIONS (Cont'n)

28. Welded joints shall not be less than 4 mm side filler welded. Undercut shall not exceed 2mm for any length of weld	Not verified
29. Other remarks	None

PURPOSE AND SCOPE OF TEST

Purpose	Performance Testing / Commercial
Date of Test	February 9, 2022
Location of Test	Brgy. Palanca, San Jose, Batangas 13° 52' 27.41"N, 121° 5' 41.06"E
Items Tested	Operator's Manual, Shredding Efficiency, Input Capacity, Speed of Components, Noise Level, Fuel Consumption Rate, and Number of Operators

METHODS OF TEST

The biomass shredder was tested based on the PNS/PAES 245:2010 Agricultural Machinery – Biomass Shredder – Methods of Test.

DESCRIPTION OF THE MACHINE

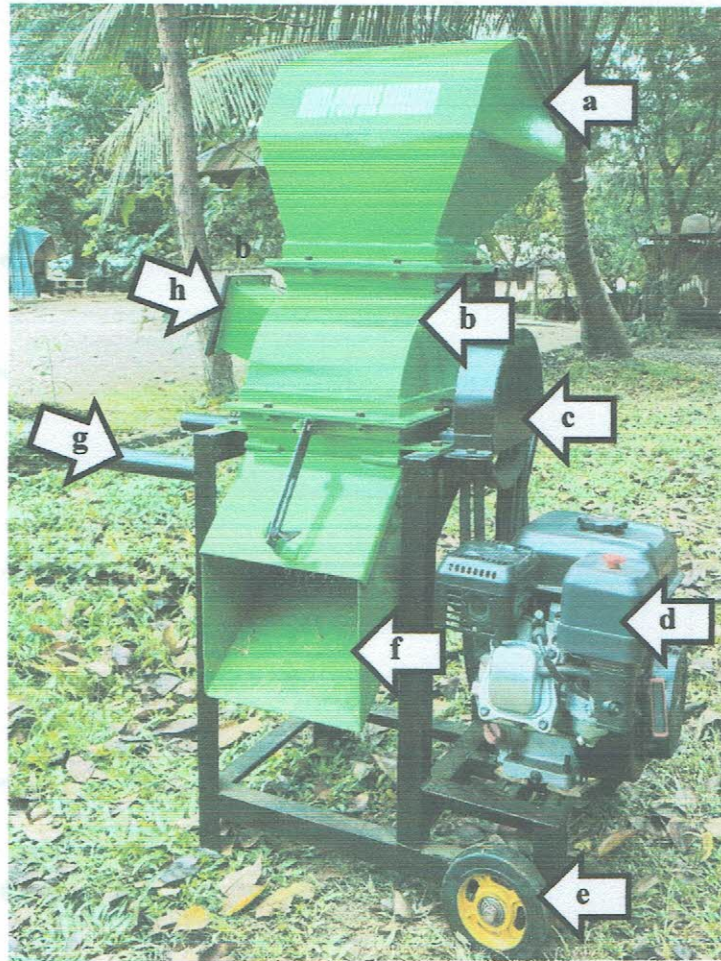


Figure 1. Parts of the RET MPS 7.5HP Biomass Shredder powered by 5.59 kW KRESS KE207 Gasoline Engine include the following: (a) Hopper, (b) Shredding chamber, (c) Belt and pulley transmission system with cover, (d) Primemover, (e) Transport wheel, (f) Outlet, (g) Transport handle, and (h) Chipping input chute.

SPECIFICATIONS

Item		Manufacturer's Specification ^a	AMTEC Verification
1	Main structure		
1.1	Overall dimensions, mm		
1.1.1	Length	1015	751
1.1.2	Width	730	1020
1.1.3	Height	1150	1153
1.2	Weight without primemover, kg	ND	NM
2	Loading hopper		
2.1	Material	MS plate	AISI 4140 ^b
2.2	Bottom opening, mm		
2.2.1	Length	250	253
2.2.2	Width	150	145
2.3	Height from the ground, mm	980	998
3	Shredder assembly		
3.1	Blade cover		
3.1.1	Type	Cylindrical	Cylindrical
3.1.2	Materials	MS plate	AISI 4140 ^b
3.2	Blade		
3.2.1	Material	Tool steel	Carbon steel ^b
3.2.2	Dimensions, mm		
3.2.2.1	Length	70	70.64
3.2.2.2	Width	25	25.04
3.2.2.3	Thickness	6.36	6.78
3.2.2.4	Shape	Rectangular	Rectangular
3.2.3	Number of blades	6-7-6-7	6-7-6-7
3.2.4	Clearance to the shredding cylinder, mm	5.44	13.12

^a The data were obtained from the operator's manual and specifications sheet provided by the requesting party.

^b Determined using Bruker S1 Titan 800 X-Ray Fluorescence Analyzer

Note: ND – No Data, NM – Not Measured

SPECIFICATIONS (Cont'n)

Item		Manufacturer's Specification ^a	AMTEC Verification
4	Output chute		
4.1	Material	MS plate	AISI 4140 ^b
4.2	Dimensions, mm		
4.2.1	Length	343	250
4.2.2	Width	250	195 (min), 406 (max)
4.3	Height from the ground, mm	290	275
5	Prime mover		
5.1	Engine		
5.1.1	Brand	KRESS	KRESS
5.1.2	Model	KE207	KE207
5.1.3	Make or manufacturer	ND	China
5.1.4	Serial number	20201101398139	20201101398139
5.1.5	Type	Four-stroke cycle, single inclined cylinder, gasoline engine	Four-stroke cycle, single inclined cylinder, gasoline engine
5.1.6	Rated power, kW	5.52	NM
5.1.7	Rated speed, rpm	3600	NM
5.1.8	Displacement, cm ³	212	NM
5.1.9	Fuel type	Gasoline	Gasoline
5.1.10	Fuel tank capacity, L	3.6	NM
5.1.11	Cooling system	Air-cooled	Air-cooled
5.1.12	Starting system	Rope recoil starter	Rope recoil starter
6	Power transmission system		
6.1	Engine to shredding cylinder	V-belt and pulley	V-belt and pulley
6.2	Engine ^c	76.2 × 2B × 19.62	76.2 × 2B × 19.10
6.3	Shredding cylinder ^c	203.2 × 2B × 25.44	203.2 × 2B × 24.95
6.4	Belt size	B-44	B-44
7	Safety devices	ND	Rubber cover for input hopper and belt cover
8	Special features	ND	Can also be used as a chipper

^a Data were obtained from the specifications sheet provided by the requesting party.

^b Determined using Bruker S1 Titan 800 X-Ray Fluorescence Analyzer

^c Pulley diameter, mm × number of belts × shaft diameter, mm

Note: ND – No Data, NM – Not Measured

RESULTS

Item		Data	
1	Condition of test material		
1.1	Test material	Chopped buko with husk	
1.2	Variety	ND	
1.3	Source	San Jose, Batangas	
1.4	Moisture content, % _{w.b.}	84.6	
1.5	Dimensions, mm		
1.5.1	Length	192.3	
1.5.2	Width	121.8	
1.5.3	Thickness	35.65	
2	Performance test		
2.1	Weight of input, kg	250.0	
2.2	Input capacity, kg/h	1115.59	
2.3	Shredding efficiency, %	94.35	
2.4	Unshredded, %	5.7	
2.5	Speed of components, rpm	Without load	With load
2.5.1	Engine shaft	3803	3734
2.5.2	Shredding shaft	1411	1359
2.6	Noise level, dB(A)	Without load	With load
2.6.1	Feeding operator	89.7	93.9
2.6.2	Output collector	90.0	94.1
2.7	Fuel consumption rate, L/h	1.57	

RESULTS (Cont'n)

Table 1. Elemental composition of the blade of RET MPS 7.5HP Biomass Shredder

Element	Percent Composition, % ^a
Iron	99.311
Silicon	0.189
Vanadium	0.060
Chromium	0.031
Manganese	0.185
Cobalt	0.102
Copper	0.0252
Molybdenum	0.049
Tin	0.040

^a Determined using the Bruker S1 Titan 800 X-Ray Fluorescence Analyzer

Table 2. Elemental composition of the shredding chamber of RET MPS 7.5HP Biomass Shredder

Element	Percent Composition, % ^a
Iron	97.768
Silicon	0.288
Chromium	0.031
Manganese	0.750
Molybdenum	0.065
Titanium	0.010

^a Determined using the Bruker S1 Titan 800 X-Ray Fluorescence Analyzer

OBSERVATIONS

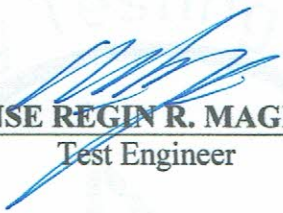


Figure 2. Operation of the RET MPS 7.5hp Biomass Shredder

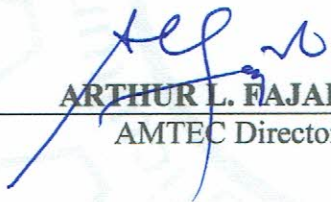
Tested by:


MARK ANTHONY R. ISUREÑA
Test Engineer

Verified by:


ALPHONSE REGIN R. MAGMANLAC
Test Engineer

Approved for Release:


ARTHUR L. FAJARDO
AMTEC Director

MAR 07 2023

Date

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